
MULALO MANWATHA

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PROFESSIONAL SUMMARY

Experienced engineering professional with over 8 years of expertise in Metallurgical, Materials, Corrosion, Welding, and Reliability engineering. I have a proven track record in power generation units, processing plants, mining operations, gas processing, and chemical plants. I have completed cross-industry projects with a quantifiable value of over \$17M, all of which have been successful. My skills include project management, reliability-centred maintenance, and planned preventive maintenance programmes. I am proficient in life cycle planning, asset management principles, and quality management, with a strong foundation in applying engineering standards and design codes such as ASME B31.8 and B31G, ASME, DIN, ISO, NACE, API, AWS, PER, and SANS. I am skilled in blueprint interpretation, corrosion protection, equipment installations, commissioning and testing, complemented by advanced knowledge of failure mode and effect analysis (FMEA), root cause analysis (RCA), and statistical data analysis. I demonstrate results-based leadership and process improvement expertise while leveraging tools like Microsoft Office, PLC and SCADA systems, Python, MATLAB, SAP PM, and SAMP to optimise operations. I am adept at utilising NDT methods for materials characterisation testing and damage assessment. My commitment to driving operational excellence is unwavering, as seen in my work with risk-based inspection (RBI) insights, risk-based outage (RBO) insights and innovative engineering solutions.

PERSONAL DETAILS

Date of Birth / Age: 32

Nationality: South African

Marital Status: Single

Gender: Male

WORK HISTORY

Welding Application Engineer, 09/2024 - Current

Afrox - Johannesburg, South Africa

- As a Welding Application Engineer at AFROX, I optimise welding processes, ensure compliance with industry standards, and drive continuous improvement across operations. My role involves technical leadership, performance analysis, and strategic business support to enhance efficiency, safety, and competitiveness.
- Conduct regular audits and performance reviews of welding processes, identifying inefficiencies in gas consumption, arc stability, and deposition rates.
- Implement optimisation strategies, including pulsed MIG and advanced shielding gas blends, to improve weld quality, productivity, and cost efficiency. This exercise reduced welding gas consumption by 7% through optimised gas flow rates and nozzle designs.
- Performed weld procedure qualification tests (WPQR) and procedure qualification records (PQR) per ISO 15614 (*welding procedure qualification*), ASME Section IX (*welding and brazing qualifications*) and SANS 10142 (*South African welding standards*).
- Implemented and monitored preventive maintenance schedules and minimised equipment downtime by 3.7% over 6 months.
- Conduct safety and quality audits to ensure compliance with OHS Act 85 of 1993, ISO 3834 (*quality requirements for fusion welding*) and AWS D1.1 (*structural welding code*).

- As a Welding Application Engineer, I provided hands-on training to over 25 technical application specialists and customer service teams on welding techniques (TIG, MIG/MAG, MMA, FCAW), equipment troubleshooting, and safety protocols. This training improved first-pass weld quality by 9% in the first 7 months, demonstrating my ability to develop and manage teams in a challenging engineering environment.
- Use metallurgical analysis (*SEM, X-ray, dye penetrant testing*) to investigate causes of weld failures, which led to cracks, porosity, and lack of fusion at AFROX facilities and client sites.
- Facilitated Lean Six Sigma projects to reduce waste from filler metal overuse and rework rates. This led to the introduction of automated gas blending systems, improving weld consistency and reducing gas waste by 12%.
- Implemented digital welding monitoring systems to track performance metrics, test and validate new welding consumables such as wires, electrodes, fluxes and gas mixtures under real-world conditions as part of a continuous improvement process.
- Advise clients on optimal base metals/filler materials for projects involving Stainless steel (316l), duplex materials, and high-strength alloys such as Inconel, Hastelloy, Aluminium, and Titanium.
- Supervise the calibration of welding machines, gas regulators, and torches according to ISO 9001 (*Quality management*) and ISO 17025 (*Laboratory testing standards*).
- Analysed and refined parameters (*voltage, amperage, travel speed, gas mix*) to reduce spatter and rework, improved deposition rates, and enhanced weld mechanical properties.
- Optimised FCAW parameters for structural steel, reducing welding time by 11%.

Reliability Engineer, 08/2022 - 08/2024

Eskom - Johannesburg, South Africa

- Implemented Reliability Centred Maintenance (RCM) strategies to transition from reactive to predictive maintenance, using tools such as vibration analysis, thermography and oil analysis in rotating equipment, reducing unexpected failures by 11% and increasing equipment uptime by 7% over 12 months.
- Performed Root Cause Analysis (RCA) on critical failures such as boiler tube leaks and turbine blade cracks using 5 Whys and Fishbone Diagrams, reducing repeat failures by 7% in 9 months.
- Established equipment criticality rankings to prioritise maintenance activities, aligning with ISO 55000 (*asset management*) standards.
- Led data collection and risk assessment for critical plant assets such as boilers, turbines, piping and electrical systems to develop risk-based inspection (RBI) plans in compliance with API 580/581, ISO 31000, and Eskom's Asset Management Policy.
- Conducted quantitative risk analysis (QRA) to prioritise inspection intervals based on Probability of Failure (Pof) and Consequence of Failure (Cof), reducing inspection costs by 6% while maintaining asset integrity.
- Optimised outage planning using risk-based outage (RBO) methodology, ensuring minimal downtime while addressing high-risk components first, resulting in an 11% reduction in forced outages.
- Developed Inspection Test Plans (ITPS) and Fitness-for-Service (FFS) assessments for ageing infrastructure, ensuring compliance with SANS 347 (*pressure equipment regulations*).
- Ensured adherence to South African regulations, including the Occupational Health and Safety Act (OHSA), pressure equipment regulations (PER) under SANS 347, ISO 9001 (*quality management*) for documentation control and ISO 14001 (*environmental management*) for sustainable asset practices
- Achieved 100% compliance in NERSA audits by standardising inspection documentation per ISO 9001.
- Trained over 23 engineers and technicians on RCM and RBI methodologies, improving plant-wide reliability practices by over 3%.
- We utilised Computerised Maintenance Management Systems (CMMS) such as SAP PM to track real-time equipment health and maintenance history.
- Developed dashboards to monitor MTBF (*Mean Time Between Failures*), MTTR (*Mean Time to Repair*), and equipment availability, improving decision-making matrix.

- Led reliability improvement projects, contributing to Eskom's Generation Recovery Plan by significantly improving plant efficiency from 67% to 83% during my tenure. This substantial increase in efficiency is a testament to the impact of my work.
- Collaborated with original equipment manufacturers (OEMS), plant managers, and engineering consultants to optimise asset lifecycle strategies.
- Spearheaded an FMEA initiative for coal-fired boilers, decreasing tube leak incidents by 8% over 16 months.
- Reason for leaving: headhunted.

Materials Engineer, 09/2020 - 07/2022

Vonmet - Johannesburg, South Africa

- Independent Metallurgical Consultancy performing condition assessments on high-energy plant equipment.
- Conducted comprehensive corrosion evaluations on pipelines, boilers, pressure vessels, and heat exchangers using ultrasonic testing (UT), radiography (RT), and electrochemical techniques to determine degradation mechanisms (pitting, stress corrosion cracking, erosion-corrosion).
- Implemented corrosion inhibition programs using chemical treatments and cathodic protection, aligning with NACE SP0169 and ISO 15589-1 standards.
- Developed a corrosion mapping system for pipelines, applying ISO 12944 (*corrosion protection of steel structures*), reducing unplanned shutdowns by 12% and extending service life by a predicted 5 years.
- Led a corrosion-resistant coating trial on pipelines and heat exchangers, testing thermal-sprayed aluminium (TSA) and high-performance epoxies per ISO 21809-3.
- Implemented graphene-enhanced coatings on pipeline joints, improving erosion resistance by over 20% compared to traditional coatings.
- Performed metallurgical failure investigations on cracked pressure vessels and heat exchangers using scanning electron microscopy (SEM), energy-dispersive X-ray spectroscopy (EDS), and hardness testing to identify root causes such as hydrogen embrittlement and fatigue.
- Reduced boiler tube failures by 15% through optimised water chemistry control and material upgrades to SA-213 T91 steel as per ASME BPVC Section II.
- Ensured all materials engineering activities adhered to South African National Standards (SANS), SANS 347 (*pressure equipment*), International Standards including ASME Boiler & Pressure Vessel Code (BPVC), API 510/570/653 (*pressure vessel & piping inspection*) and Occupational Health & Safety: OHS Act 85 of 1993.
- Achieved zero non-conformances in 3 consecutive ISO 9001 audits by standardising material certification and inspection reports.
- Introduced non-destructive testing (NDT) protocols (PT, MT, TOFD) for early defect detection, reducing catastrophic failure risks by an estimated 7% over 6 months.
- Reason for leaving: better job opportunity.

Metallurgical Engineer, 06/2018 - 08/2020

Eskom - Johannesburg, South Africa

- Led metallurgical data collection for RBI programs on critical power plant components, including Boiler tubes, Steam turbines, Pressure vessels and high-energy piping systems
- Implemented RBI for 6 coal-fired units, reducing inspection costs by 9% in 8 months while maintaining 100% coverage of high-risk components
- Established RBO prioritisation matrices that reduced forced outage durations by 7% through optimised metallurgical inspection scheduling
- Conducted 115 failure analyses on power plant components using Metallography (*optical/SEM microscopy*), hardness testing (*Vickers, Rockwell*), fractography (*ductile vs. brittle fracture analysis*) and chemical analysis (OES, EDS).

- Implemented plant-wide corrosion monitoring programs including Online corrosion rate monitoring (LPR probes), deposit analysis (XRD, XRF) and water chemistry controls (per EPRI guidelines). This led to, amongst others, the identification of stress corrosion cracking in low-pressure (LP) turbine blades, leading to a material upgrade to 12% Cr steel as per ASTM A565.
- Developed failure prevention guidelines that reduced repeat boiler tube failures by 11% across 3 power stations
- Performed remaining life assessments for ageing components using creep life prediction, wall thickness measurements (*UT scanning*) and microstructural degradation (*carbide precipitation, spheroidization*)
- Extended service life of 4 boiler drums by an estimated 3 years through weld repair procedures (as per ASME Section IX), post-weld heat treatment optimisation and revised operating parameters
- Ensured all metallurgical engineering activities complied with Eskom Technical Specifications (ETS 08-1, ETS 08-4), South African National Standards (SANS 347, SANS 10208), international codes such as ASME Boiler & Pressure Vessel Code, API 571 (*damage mechanisms*) and ISO 15608 (*welding qualifications*).
- Reason for leaving: better career prospects.

Metallurgical Graduate Engineer, 10/2017 - 05/2018

Glencore - Johannesburg, South Africa

- Conducted research project work on Plant Process improvement under the Research, Development, and Testing (RT&D) department. The research project included: conducting daily metallurgical accounting (*mass balances, recovery calculations*) to track chrome concentrate quality, recovery trends and recovery efficiency, sampling and testing of feed, concentrate, and tailings to ensure process stability, monitoring flotation, gravity separation, and spiral concentrator performance, identifying deviations and recommending corrective actions.
- Ensure QA/QC compliance in laboratory testing, following ISO/IEC 17025 (General requirements for testing laboratories).
- Adhere to Glencore's Safety, Health, and Environmental (SHE) policies, including the Mine Health and Safety Act (MHSA) of South Africa and the Occupational Health and Safety Act (OHS Act 85 of 1993)
- I assisted Metallurgists in root cause analysis (RCA) of process upsets, such as low recovery and high tailings losses.
- Work closely with plant operators, senior metallurgists, and maintenance teams to implement best practices.
- Attended training programs on process optimisation, safety, and industry standards.
- Reason for leaving: completed training program.

SKILLS

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| • Project Management | • Understanding of engineering standards and regulations |
| • Reliability Centred Maintenance | • Blueprint interpretation |
| • Planned Preventive Maintenance (PPM) | • Equipment commissioning and testing |
| • Contracts management | • Failure Mode and Effect Analysis (FMEA) |
| • Corrosion protection | • Root Cause Analysis (RCA) |
| • Material science expertise | • Statistical data analysis |
| • Metallurgical expertise | • Technical writing |
| • Process improvement expertise. | • Results-driven Leadership |
| • Quality Management expertise | |

TECHNICAL SKILLS

- Microsoft Office (Word, Excel, Outlook, PowerPoint), Python, MATLAB, ASPEN, SCADA, PLC, Smart Access, SAMP and Knowledge of SAP
- Knowledge of ASME B31.8 ,314 & B31G
- Application of design codes: ASME, DIN, ISO, NACE, API, AWS, PER, and SANS
- NDT Methods: UT, RT, PT, MT, PEC (Pulsed Eddy Current)
- Materials characterisation: SEM/EDS, XRD, Hardness Testing
- Corrosion Testing: LPR, EIS, Potentiodynamic Polarisation
- Materials characterisation: SEM/EDS, XRD, Metallography
- Damage Assessment: Creep, Fatigue, Corrosion Mechanisms

EDUCATION

Bachelor of Engineering Honours (BEng Hons): Welding Engineering, 11/2025

University of Pretoria

- Subjects: Fabrication Engineering, Welding Metallurgy , Welding Processes , and Design of Welded Structure (Pass with Distinction), Research Project

Bachelor of Engineering: Metallurgical Engineering, 01/2017

University of Pretoria

National Senior Certificate: Grade 12, 01/2011

Tshikevha Christian School

CERTIFICATIONS

- Registered Professional Engineer, Engineering Council of South Africa (ECSA), 2025
- International Welding Engineer (IWE), International Institute of Welding (IIW), 2025
- NDT Visual Testing Level II, School of Applied Non-Destructive Examination, 2020
- ISO 14001:2015, ISO 45001:2018 and ISO 9001:2015 accreditation

KEY PROJECTS COMPLETED

- Pipeline Corrosion Mitigation, Materials/Corrosion Engineer, 5-year life extension, \$3.2M
- Boiler Tube Life Extension, Reliability/Metallurgist, 20% fewer outages, \$3.8M
- Welding Optimisation, Welding Applications Engineer, 30% faster repairs, \$1.5M
- HIC Prevention in Sour Gas, Materials Engineer, Zero HIC failures, \$2.5M
- Turbine Blade Reliability, Metallurgist, 35% higher MTBF, \$4.1M

PLANT/EQUIPMENT KNOWLEDGE

- Coal-Fired Power Stations, Processing Plant, Mining Operations, Gas Plant, Chemical Plant
- Heat exchanger, Boiler, Steam Turbines, Fans, Pumps, Rotary Kiln, Electric Arc Furnace, Gas and Arc Welding equipment, Precipitators, Feedwater pumps, etc.

REFERENCES

- Michael Mkhize, Chief Engineer, Eskom, mkhizemm@eskom.co.za, +2772 477 8279
- Gerard von dem Bongart, Manager, Vonmet, gerardvdb@vonmet.co.za, +27827846126